Intersection between surfaces using Computer Extended Descriptive Geometry (CeDG): application to the focal illumination of a sphere

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Abstract

Descriptive Geometry (DG) and its derived representation systems, such as dihedral system, have defined the fundamentals of graphical representation in engineering since the second half of the 19th century, establishing an important scientific-technical corpus. This one includes many texts that continue to provide a solid basis for university teaching, despite the fact that their use in the professional field has been practically replaced by computer aided design software (CAD) [1]. CAD tools share the use of methodologies derived from constructive solid geometry (CSG) and boundary-based representation (B-Rep), combined with different strategies. The geometry of CAD model is mainly represented by fit curves and surfaces, such as B-splines, which provide high control and accuracy.

Computer Extended Descriptive Geometry (CeDG) is a new approach to computer modelling of 3D geometric systems [2], which tries to address the limitations of current CAD systems pointed out by some authors [2-4]. Briefly, CAD tools allow the construction of virtual prototypes of 3D systems that can be manipulated in space and easily projected according to the chosen representation system. However, they do not facilitate the creation of the model when it depends on some implicit parameter, in addition to presenting shortcomings in the calculation of flat patterns of sheet metal surfaces.

The CeDG approach combines the ability of DG to solve spatial geometric problems with the ability of dynamic geometry in the process of building geometric-algebraic models. Unlike CAD systems, CeDG parametric models preserve the integrity of curves and surfaces. A preliminary version of the CeDG approach implemented on the dynamic geometry software Geogebra has demonstrated its ability to overcome some of the above limitations [2,5].

The objective of this study is to analyze the methodology used in CeDG models to obtain the intersection curve between two surfaces. The usual procedure in DG consists of defining it by means of its projections, which are calculated in turn by interpolation on a set of points belonging to them. The points are obtained by means of DG techniques, applied iteratively. Accordingly, the accuracy and complexity of the resulting curve is proportional to the number of points obtained [6]. The CeDG approach uses an extension of the DG procedures, which allows the generation of a specific locus function for each projection of the intersection curve sought. The locus function is automatically determined from the sequence of geometric-algebraic instructions associated with the calculation of a single generic point of the intersection [2].

This methodology is evaluated by calculating the curve bounding the surface of a sphere that is illuminated by a focal (spot) light beam, using the usual DG technique and CeDG approach. Both cases were implemented in Geogebra in order to avoid any further differences between the solutions. The system was parameterised to control the position and solid angle of the light cone, the radius of the sphere and its position in relation to the light spot.

The solution curves obtained are shown in Figure 1, resulting from the composition of part of the cone-sphere intersection with an arc of the separatrix circle associated to the given focus. The intersection curve obtained by means of CeDG got the maximum precision defined by default in Geogebra, as it was expected since it did not need the approximation by interpolation. To achieve this result, advantage was taken of the particular position of the system, composed of two quadrics of revolution with axes parallel to the vertical plane of projection, for which the vertical projection of the cone-sphere intersection corresponds to a hyperbola [7, Chapter 30].

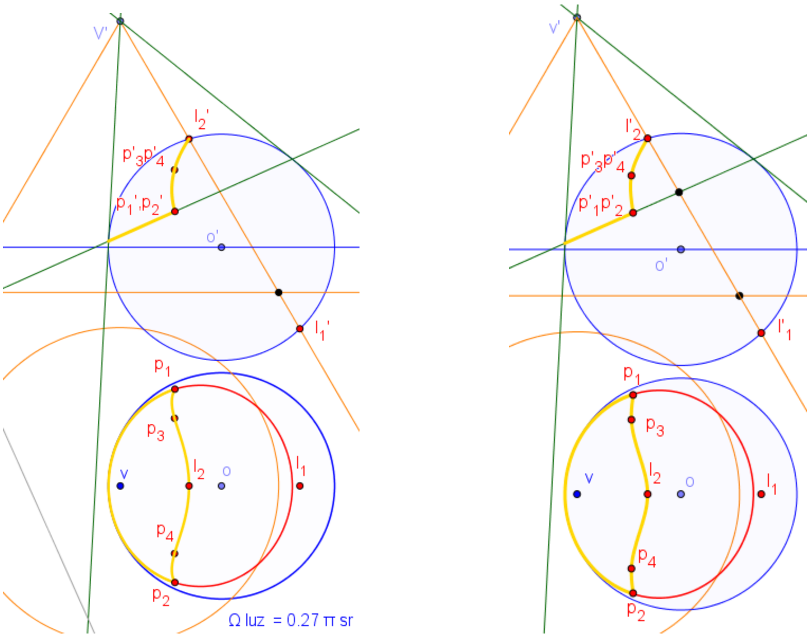

Figure 1: System projections obtained by CeDG (left) and by classical DG (right). The illuminated surface of the sphere is limited by its perimeter curve (yellow).

The accuracy of the solution obtained by the usual DG method was lower and dependent on the system parameters. The difference of eccentricities between the hyperbola associated to each intersection curve and the exact hyperbola was taken as a metric related to the accuracy of the solutions found.

The results of this study allow us to conclude that the CeDG methodology for the calculation of intersections between surfaces outperforms the classical DG methodology. Although the work is limited to the intersection between quadrics used, other published results suggest that this advantage of CeDG is extrapolable also to transformed curves linked to plane patterns (sheet metal) [2]. CeDG also outperformed CAD in this task in the sheetmetal domain [5].

Additionally, any modification in the projection direction or in the overall dimensions of the geometric system is automatically propagated to the locus functions which define the intersection curve in the CeDG model, avoiding the need to obtain the points that belong to the contour lines, which are required in the classical DG methodology.

References

1. Weisberg, D.E. Computer-Aided Design Strong Roots at MIT. In *The Engineering Design Revolution:The People, Companies and Computer Systems That Changed Forever the Practice of Engineering*; http://cadhistory.net/: 2008; p. 650.
2. Prado-Velasco, M.; Ortíz Marín, R.; García, L.; Rio-Cidoncha, M.G.D. Graphical Modelling with Computer Extended Descriptive Geometry (CeDG): Description and Comparison with CAD. *Computer-Aided Design and Applications* **2021**, *18*, 272-284, doi:10.14733/cadaps.2021.272-284.
3. Migliari, R. Descriptive Geometry: From its Past to its Future. *Nexus Network Journal* **2012**, *14*, 555-571, doi:10.1007/s00004-012-0127-3.
4. Stachel, H. The Status of Todays Descriptive Geometry Related Education (Cad/Cg/Dg) in Europe. *Journal of Graphic Science of Japan* **2007**, *41*, 15-20, doi:10.5989/jsgs.41.Supplement1_15.
5. Prado-Velasco, M.; Ortiz-Marín, R. Comparison of Computer Extended Descriptive Geometry (CeDG) with CAD in the Modeling of Sheet Metal Patterns. *Symmetry* **2021**, *13*, 685.
6. Leighton Wellman, B. *Geometría Descriptiva [Technical Descriptive Geometry]*; Editorial Reverté, S.A.: Sevilla, 1987; p. 615.
7. Taibo, Á. *Geometría descriptiva y sus aplicaciones II: Curvas y superficies* Editorial Tebar S.L.: Madrid, 2009; Volume II, p. 446.